SECTION 15182

SHOP FABRICATED STAINLESS STEEL TANKS

PART 1 - GENERAL

1.1 SUMMARY

A. This Section specifies the requirements for the design, manufacture, testing, delivery and installation of stainless steel tanks complete with all accessories.

1.2 RELATED WORK

A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the referenced tanks:
   1. Tanks listed in Section 15351.01_dat, “High Resistivity Water (HI-RES) System Data Sheets.”
   2. Reference Drawings attached at the end of this section.
   3. Drawing B51.I3.201A for the Process and Information Diagram of the HI-RES tank.
   4. Drawing B51.EQ4.01 for preliminary tank configurations.

1.3 SUBMITTALS

A. Provide the following with the Bid:
   1. Preliminary drawings of the tanks.
   2. Drawings to include; fabrication and test codes, dimensions, nozzle connections, manhole locations and bill of materials.
   3. At the time of bid, alternates should be submitted along with the original bid.

B. Provide the following within two (2) weeks of award of Contract and prior to start of fabrication:
   1. Certified Shop Drawings showing fabrication and test codes, detailed dimensions, nozzle locations, manhole locations, support structure, miscellaneous details and complete bill of materials.

1.4 QUALITY ASSURANCE

A. It is the responsibility of the Manufacturer to perform detailed design, provide construction details, and comply with all appropriate laws, ordinances and regulations controlling fabrication of storage tanks.

B. All tank design shall be performed by a Professional Engineer Licensed in the State where tank will be installed. All calculations and drawings shall be sealed by the design engineer. Sealed calculations including venting and emergency venting shall be submitted to the Owner and the municipal code enforcement authorities. Start of fabrication does not begin until the calculations have been submitted.

C. Review of submittals by Owner does not relieve the tank fabricator of his responsibility for complete design of each tank, including final materials selection and application.

D. Inspect all materials prior to shipment from factory for defects and repair or replace all defective components prior to shipment.

E. All fabrication shall be done by experienced personnel using appropriate methods and safety procedures.

F. Actual locations of nozzles, connections, and openings shall be indicated on supplier's submitted drawings.

G. Manufacturer shall have the capability to provide repair, maintenance, and parts supply service for each tank and all appurtenances.

H. Industry Standards: Design and fabrication shall be in accordance with the state and local codes, ordinances, and regulations of Boise, Idaho and the latest edition of codes and standards of the following, where applicable.

1. American National Standards Institute (ANSI).
2. American Society of Mechanical Engineers (ASME).
3. American Society for Testing and Materials (ASTM).
4. American Welding Society (AWS).
5. Steel Structures Painting Council (SSPC).
6. Occupational Safety and Health Administration (OSHA).
7. American Petroleum Institute (API).
8. Underwriters Laboratory (UL)
9. 2012 International Building Codes including all attachments, amendments and revisions.

I. If Specification requires compliance with two or more industry standards which establish different or conflicting requirements, confer with Owner's representative before proceeding with affected operations.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Cleaning:

1. Perform final cleaning of all internal and external surfaces after all manufacturing and testing operations have been completed.
2. Dry each tank / vessel and ensure all interior surfaces are free of foreign materials.
3. Cleaning procedures are the Supplier's responsibility.

B. Deliver, store, and handle products/equipment in accordance with the following:

1. Use methods which prevent damage, deterioration, and other loss during shipping and temporary on-site storage.
2. Plug, cap, or cover with plywood (or other suitable material) all exposed openings.

3. Furnish instructions for proper unpacking.

C. Convey products to Project site by means which minimize risk of loss and deterioration during transit.

D. Deliver separate loose parts boxed in a plywood container and ship as a unit with the equipment.

E. Identify all shipping boxes by the Purchase Order number, Owner's name, Project name, Project location, equipment number, and description of the contents with ink, paint, or other indelible material markings. Securely affix a copy of the completed packing list to each separate shipping package.

1.6 LOADINGS

A. Tank design pressure shall be 2.5 psig above maximum liquid level.

1.7 WARRANTY

A. Warrant each tank and all materials and appurtenances will be free of defects in materials, design, fabrication and installation for a minimum period of one year after tanks are placed in service, or 18 months after delivery, whichever occurs first.

1.8 APPROVED MANUFACTURERS

A. Hobson, Boise ID

B. YMC, Boise ID

C. Diversified Metal Products, Boise, ID

D. Modern Welding, Houston, TX

E. Caldwell Tanks, Louisville, KY

F. Palmer Tanks, Garden City, KS

G. Edlon

H. B. H. Tank, Inc., Los Angeles, CA

I. Owner approved equal.

PART 2 - PRODUCTS

2.1 SOURCE QUALITY CONTROL

A. Testing/Inspections:

1. Comply fully with all requirements, test, inspections, reports, documentation, and approvals contained in this Section and/or standards implemented by this Specification. Make available on request a copy of any part of the Supplier's quality assurance records. The Supplier's tests, inspections, reports, and documentation, as well as those of his subcontractors, are subject to review and witnessing by the Owner's Representative.
2. Notify the Owner not less than ten working days prior to the start of fabrication.
3. Maintain records of all inspections and tests. The records are to indicate the nature and number of deficiencies found, the quantities approved and rejected, and the nature of corrective actions taken. Forward copies of the inspection and test data to the Owner's Representative prior to or at time of shipment.
4. Notify the Owner's Representative not less than five working days prior to conducting inspection or testing required by this Specification and/or standards implemented by this Specification.
5. Reviews or inspections by the Owner's Representative in no way relieve the vendor of responsibility to comply with this specification.

2.2 MATERIALS

A. Fabricate each tank from stainless steel, Type 304 or 304L, ASTM A240.

B. Provide material test reports for all plate, sheet, piping, tubing, flanges, and fittings in contact with the process fluid with the exception of items of standard manufacture.

C. Purchase stainless steel materials iron-free. Maintain this condition during all phases of fabrication. Follow the recommendations of Section 8 of ASTM A 380 for prevention of contamination.

D. Unless otherwise specified, passivate interiors of tanks before shipment.

2.3 EQUIPMENT

A. General:

1. Conform to attached Reference Drawings and Section 15351.01_dat, “High Resistivity Water (HI-RES) System Data Sheets.” The B51 HI-RES tank is to be a mirrored copy of what is indicated in the Reference Drawings.
2. Material Thickness: The thickness of tank sides, top, and bottom to be determined for atmospheric tanks in accordance with design standards per the Reference Drawings, but in no case less than 1/4 inch.

B. Externals:

1. Dissimilar Materials: Unless otherwise specified, clips, supporting skirts, structural shapes, brackets, etc., welded to the outside of a stainless steel tank for support of insulation etc., may be carbon steel. A 300 series stainless steel pad of thickness equal to the adjacent tank wall shall be welded to the tank and the carbon steel attachment welded to the pad.

   a. Clips may be of Type 304 stainless steel.

b. Surface of stainless steel plate is not to be contaminated with carbon steel. Use either separate equipment for rolling plate and forming beads or thoroughly clean and cover equipment after working carbon steel beforehand on the same equipment.

2. Tank Support Design:

a. Tanks to be self-supporting.
b. If vessel supports are specified, but no details given, design, fabricate, and guarantee the supports.
c. For vertical leg supported tanks, unless otherwise specified, the tank supports shall be designed for a 4'-0" height from the low-point of the tank bottom to the supporting surface.

3. Tank Roof Design:

a. Tank top or roof type shall be per the Reference Drawings.
b. Design tank roof loading for maintenance and personnel equipment loads. Roof load shall be capable of supporting 225 lb. individual with 50 lb. of equipment.

4. External Attachments: Provide all the necessary clips or brackets for attachment of ladders, platforms, lifting lugs, anchor clips, and other accessories indicated in Section 15351.01_dat, "High Resistivity Water (HI-RES) System Data Sheets" and Reference Drawings.

a. All external attachments, such as ladders, platforms, etc., furnished under this Specification, shall be designed to comply with the latest OSHA regulations.
b. Supply all caged ladders and access platforms with handrails where shown on the Reference Drawings. These items shall be galvanized steel construction.
c. The Tank Vendor shall be responsible for the complete installation of the external attachments required per the Reference Drawings including any field assembly.

5. Nozzles: Nozzles to be of the size and number, as called for on the Reference Drawings.

a. Nozzle Necks, Dip Pipes: Schedule 40S stainless steel, ASTM A312 TP304.
b. Nozzle Connections:

1) Full threaded couplings, class 2000, ASTM A182 Grade F304 or F304L, ANSI B16.11.
2) Flanged, class 150, ASTM A182 Grade F304 or F304L, ANSI B16.5, welding neck.
3) Connections for level transmitters shall be flanged regardless of size.

c. Flanged nozzles shall have a nominal projection of 6 inches unless otherwise indicated on Reference Drawings.
d. Blind flanges and manhole covers shall be constructed of stainless steel, Type 304.
e. Nozzle Orientation: Owner's Representative reserves the right to locate and orient nozzles when approval Drawings are submitted at no increase in price unless the quantity and/or size(s) change.

6. Gaskets: Supply gaskets necessary for tests and shipment. Supply two (2) sets of service gaskets required for blind flanges, manholes, etc. Gaskets material shall be per the Reference Drawings. The Tank Vendor shall offer an alternate suggested gasket for the indicated service.

7. Bolting:

   a. External and internal bolts and studs shall conform to ASTM A 193, Grade B8, and nuts to ASTM A 194, Grade 2H, unless shown otherwise on the Reference Drawings.
   b. External bolts and nuts used for platforms, ladders, and other external attachments shall conform to ASTM A 307, Grade A, and ASTM A 563, Grade B, heavy hex.
   c. Stamp all studs, bolts, and nuts with the manufacturer's marking to indicate they are special steel (i.e., B8, B7, etc.).

8. Instrumentation: Refer to the Process and Information Diagrams and Reference Drawings for Level Instrumentation requirements. The TankInstallation Contractor shall be responsible for furnishing and the complete mechanical mounting of the level instrumentation including any field assembly required. Wiring will be by others.

C. Internals:

1. All internals to be of the same alloy as the tank unless stated otherwise on the Data Sheets.
2. Tank internals include dip pipes, support brackets, suction pipes, baffle plates, structural shapes, etc., as shown on the Reference Drawings.
3. The Tank Vendor shall be responsible for the complete installation of the internal attachments required per the Reference Drawings and Section 15351.01_dat, “High Resistivity Water (HI-RES) System Data Sheets,” including any field assembly.

2.4 FABRICATION/SHOP ASSEMBLY

A. Welding:

1. Adhere to AWS A2.4 for all welding and nondestructive test symbols. Include symbols on all fabrication detail drawings.
2. Adhere to AWS A3.0, Terms and Conditions, for definitions of welding processes, methods, techniques, and applications.
3. Unless otherwise specified on the Reference Drawings, the welding process shall be by the TIG procedure, using Argon as the inert gas on tank interior, and SMAW on tank exterior.
4. Use austenitic stainless steel wire brushes on stainless steel welds.
5. All pipe, plate, and structural butt joint welds are to be continuous, full penetration.
6. All pipe and nozzles must be installed using full penetration welds. Nozzles in roofs of atmospheric tanks may be welded using double fillet welds (inside and outside).
7. Backing rings shall not be used.

8. All nozzles, fittings, and couplings 2 inches diameter and smaller shall be Gas Tungsten Arc welded (GTAW).
9. All welding shall be performed by welders certified in accordance with applicable provisions of Section IX of ASME BPV code.

B. Tolerances:

1. Nozzles: Elevation and lateral location of nozzle center plus or minus 1/8 inch for nozzles smaller than 8 inches and plus or minus 1/4 inch for nozzles 8 inches and larger. Location of nozzle face from dimensioned point shall be plus or minus 1/16 inch for nozzles smaller than 8 inches and plus or minus 1/8 inch for nozzles 8 inches and larger.
2. Nozzle faces shall not deviate from normal plane more than following tolerances:

Size (in.)	Tolerance (in.)
0 – 18	1/32
20 and larger	1/16
Manholes	1/16

3. Support lugs and brackets shall not deviate more than 1/8 inch in elevation and lateral location.
4. Warpage to be corrected by rework or machinery.

C. Finish:

1. Finish vessel interior surface, including welds, as specified on the attached Reference Drawings. Vessel finishes are defined by ASTM A 480 or by the Roughness Average (Ra).

D. Equipment Tagging:

1. All material furnished shall include the equipment number per Drawing B51.I3.201A and Section 15351.01_dat, “High Resistivity Water (HI-RES) System Data Sheets.” All submittals, correspondence, Drawings, invoices, and other communication will also bear this equipment number in addition to Purchase Order numbers, Owner's name, project name, and project location. Where equipment fabrication is of alloys not compatible with stainless steel, fabricate the tag from the equipment material.
2. For each tank, provide permanently attached, conspicuously displayed stainless steel nameplate inscribed with following minimum information:

   a. Name of manufacturer.
   b. Design pressure and temperature.
   c. Manufacturer's serial number.
   d. Year built.
   e. Equipment number.
   f. Tank height.
   g. Tank dimensions.
   h. Owner's purchase order number.
   i. Manufacturer's purchase order number.
   j. Designated chemical contents.

Also refer to Reference Drawings for Code Stamp requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Perform all required field assembly and supervision to complete installation.

B. Tighten all bolts to maximum tightness without overstressing or breaking metal. Lubricate with an anti-seize compound before tightening.

C. Attach and secure anchor lugs and secure to foundation.

D. Attach, brace and secure each access ladder (if applicable).

E. Install and secure each level gauge assembly with accessories and connecting piping.

3.2 FIELD QUALITY CONTROL

A. It shall be the responsibility of the tank manufacturer and erector to ensure that all materials are as specified and that each tank is erected and installed properly and in accordance with the Drawings and manufacturer instructions.

B. Test each tank in place by sealing connections, filling with water to 6 feet (2.5 psig) over the overflow level, allowing water to stand for a minimum of 4 hours. Notify Owner of test a minimum of 5 days in advance so that test may be witnessed. Record field test information and submit to Owner. When test is complete, dispose of water in a safe manner, as directed by the Owner.

C. Provide opportunity for owner to inspect tanks at any stage of shop fabrication and erection.

D. Correct all defects disclosed by inspections and tests and perform all additional work necessary to provide secure, completely functional installation.

E. Clean each tank and work area of all trash, dirt and equipment when work is completed. Flush out tank interior with water and flush out all connections before connecting to piping system and placing in service.

3.3 ATTACHMENTS

A. The following Reference Drawings are attached to and made part of this Section.

NOTES:

1. INTCRPRET THI S DRAVING JN ACCORDANCE V I TH ANS!/ ASME Yl4 .! 00- 2004
2. USE QUAL! f"IEO VE L OERS AND VEL OING PROCEDURE SPEC:FJCATIDNS IN ACCORDANCE ı.t!TH ASME Bi P V CODE SECH ON IX .
3. ALL UNITS ARE U.S. CUSTOMARY.
4. BUTT VEL D SE AM AT FABRICATOR' S DISCRETION.
5. PRIOR TO PLACI NG TANK I NTO SERVI CE, REMOVE AL L MIL L SCALE , \JEL DING SL AG, AND VEL D SPATTER FROM ! ANK I NTERIOR SURFACES
6. REPAD DIAMCTER IS NOMINAL CUT AS REQUIRED TO FIT AROUND TUBE SU PORTS.
7 EQUIPMENT TAG NO. AND 0 \JNER' S P.O. NO ARE TO BE DETERMINED BY 0 \JNER.
8. MANUFACTURER'S SERI AL NO. AND P.O. ND ARE TD BE DETERMINED BY MANUF CTURER

3

HANUFACTURER NAME• DIVE'.RSIFIED METAL PRODUCTS, INC
DESIGN PRESSURE• 2.5 PSIG ABOVE MAX LI QUI D LEVEL
DESIGN TEMPERATURE, 75•F
MANUFACTURER'S SERIAL NO,
DATE OF MANUFACTURE• 2016
EQUIPMENT TAG NO.•
EQUIPMENT NAME• PC\o/ <HIRES> RECIRCULATION TANK
TANK HEIGHT• 9• - 0·
TANK LENGTH X 'wIDTH• 10'- 0' X 10'-0'
O'wNER'S P.O. NO.,
MANUF ACTURERS P.O. NO,
DESIGNATED CONTENT•S HI GH- RESISTIVIT Y \o/ATER

NAMEPLATE DETAIL

\OZ	SIZE	SCHED/IJALL	RATING	FACING	SERV I CE/ REMI>RKS
NI 7	2·	XS	LSO•	RFSD	AIR BUBBLE RELEASE CONNECTION
NI&	2·	XS	L5 0io	RFSO	SI GHT GL ASS
N15	2 ·	XS	150•	RFSO	SI GHT GLASS
NI 4	2 '	XS	150•	RFSO	NITRO GEN BLANKET SUPPL Y
NI 3	2·	XS	I 50 tt	RF'SO	H! - RES, \JA TE;R FROM CUPON
NI2	2'	XS	150 #	RFSO	SP ARE
Nil	2'	XS	150#	RFS☐	NON POTABLE IJATER
NIO	2·	XS	150#	RFSO	II P\J V t>TER SUPPL Y
N9	4'	XS	150#	RFSO	HH S. L L LE VEL AL ARM,
NS	4	XS	150!	RFSO	<JL TRASONIC L EVEL SENSOR
N7	4'	XS	150#	RFSO	DRAIN
N6	4 •	XS	150ıı	RFSO	OVERF LL☐'J
N5	4'	XS	150#	FS☐	SPARE
N4	6·	XS	150#	RFSD	SPARE
N3	20'	XS	! 50	RFSO	HI - RES RETURN
N2	20·	XS	! 5 0	RFSO	MAINTENANCE ACCESS
NI	2 4'	XS	!SOM	RFSO	HI - RES SUPPL Y

QTY Rm	000.A11RE DR DESCRIPTION	PAI!T/PART NJ	rIND Nil
AR	PLATE, 3/16' THICK 304/304L STAINLESS STEEL, ASTM A240	NAMEPLA TE	23
AR	ANGLE, 3' X 3' X 1/2' THK 304/ 304L STAINLESS STEEL ASTM A276	TIE DCJ',/N LUGS	22
I	GASKET, 20', 150 LB, 1/8' THK, \o/ FL E XIBLE GRAPHITE SPIRAL \o/0\JND, 304 STAINLESS STEEL, ASHE B1620	N2 GASKET	21
20	NUT, HEAVY HEX, I 1/8'-7 UNC-2B TEFLON COATED CARBON STEEL , ASTH A194-2H	N2 NUTS	20
20	STUD/BOLT, I 1/ 8' - 7 UNC-2A X 6' LONG ZINC COATED CARBON STEEL , ASTM A193-B7	N2 BOLT S	19
AR	TUBE, 3' X 3' X 3/16' THK 304/304L STAI NLESS STEEL, ASTM A554	SUPPORTS	18
AR	PIPE, 2' NPS, SCHEDULE 8 0 <XS), SHLS 304/304L STAINLESS STEEL, ASTM A312	NOZZLE NIO-N 17 NEC	17
AR	PIPE, 4• NPS SCHEDULE 8 0 <XS>, SMLS 304/304L STAINLESS STEEL, ASTM A312	NOZZLE N5- N9 NECK	16
AR	PIP E, 6' NPS, SCHEDULE 80 <XS), SMLS 3 04/ 304L STAINLE'.SS STEE.L ASTM A312	NOZZLE N4 NECK	15
AR	PIPE, 20' NPS, SCHEDULE XS, SMLS 304/304L STAINL ESS STEEL, ASTM A312	NOZZLE N3 NECK	14
AR	PIPE, 20 ' NPS, SCHEDULE XS, SMLS 304/304L STAINL ESS STEEL, ASTM A3I2	NOZZLE N2 NECK	13
AR	PIPE, 24' NPS SCHEDULE XS, SMLS 304/ 304L STAINLESS STEEL, ASTM A312	NOZZLE NI NECK	12
AR	SHEET, 20 GAUGE 304/ 304L STAINL ESS STEEL, ASTM A240	SIGHT GLASS MOUNTS	11
AR	PLAT E 1/4' THICK 304/ 304L STAINLESS STEE'.L, ASTM A240	SHELL/ REPADS/ BAFFLE	10
AR	PLATE 5/16' THICK 304/304L STAINL ESS STEEL, ASTM A240	SHELL	9
AR	PLATE, I' THICK 304/304L STAINL E'.SS STEEL, ASTM A240	LIfTING LUGS	8
8	FLANGE, 2' NPS, 150 LB, RFSO 304/304L STAINLESS STEEL, ASTM A182, ANSI B16.S	N!O-NI 7 FL ANGE	7
5	FLANGE 4' NPS, 150 LB, RFSO 304/304L STAINLESS STEE'.L, ASTM A182, ANSI B16,5	N5 - N9 FLANGE	6
1	FLANGE, 6' NPS, ISO LB, RFSO 304/304L STAINL ESS STEEL, ASTM A182, ANSI B16.5	N4 FL ANGE	5
1	FLANGE, 20' NPS 150 LB, RFSO 304/304L STAINLESS STEEL, ASTM A182, ANSI B16.S	N3 FLANGE	4
I	FL ANGE, BLIND, 20' NPS, 150 LB 304/304L STAINLESS STEEL, ASTM A182, ANSI B16.S	N2 BLIND FLANGE	3
1	FLANGE, 20' NPS, !SO LB, RFS☐ 304 / 304L STAINL ESS STEE'.L, ASTM A182, ANSI B16.S	N2 FLANGE	2
1	FLANGE, 24' NPS, 150 LB, RFSO 304/304L STAINL ESS STEE, L ASTM A182, ANSI B16,5	NI FLANGE	I

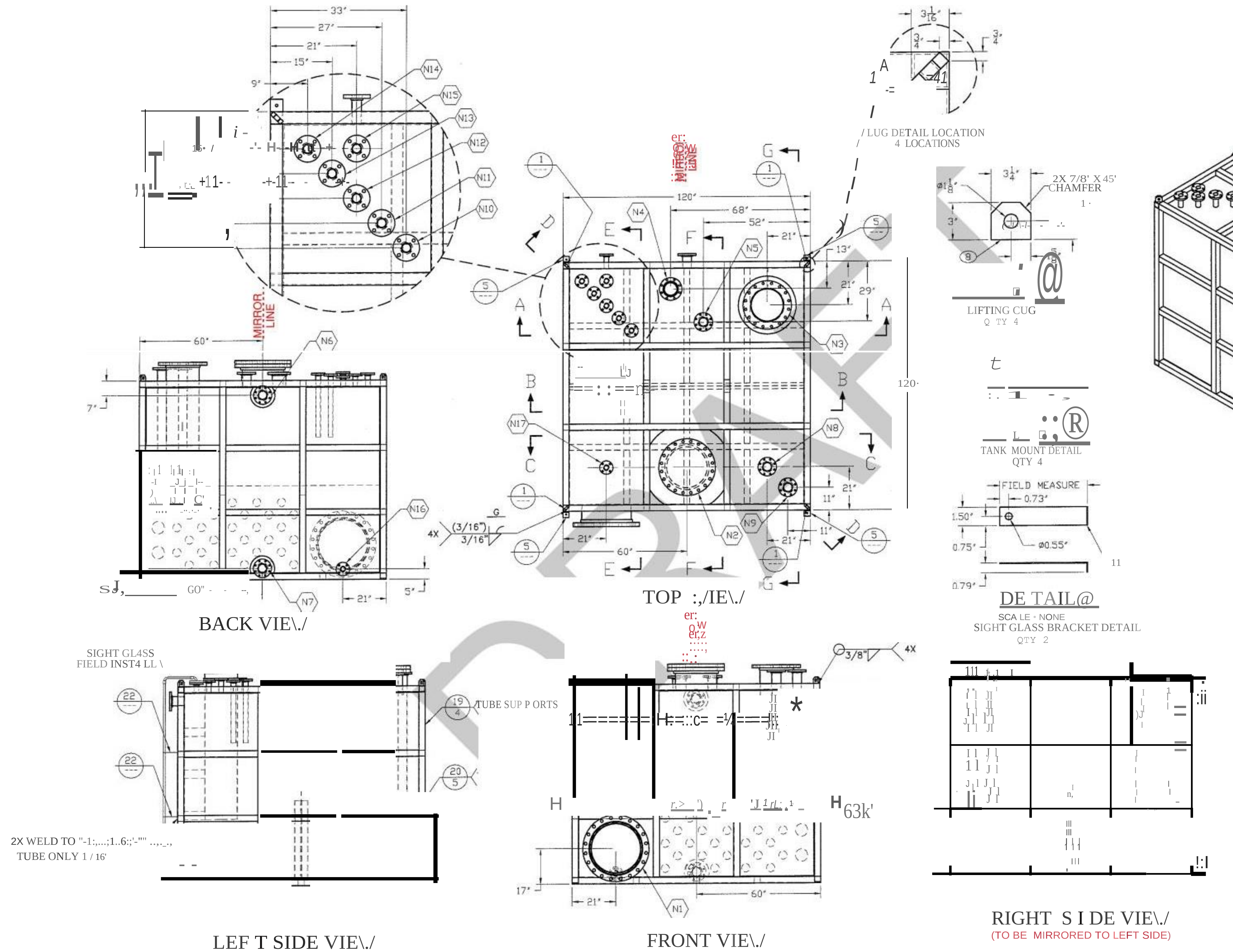
DETAILS ARE FOR REFERENCE ONLY. THE FOLLOWING DETAILS ARE FOR THE CONSTRUCTION OF

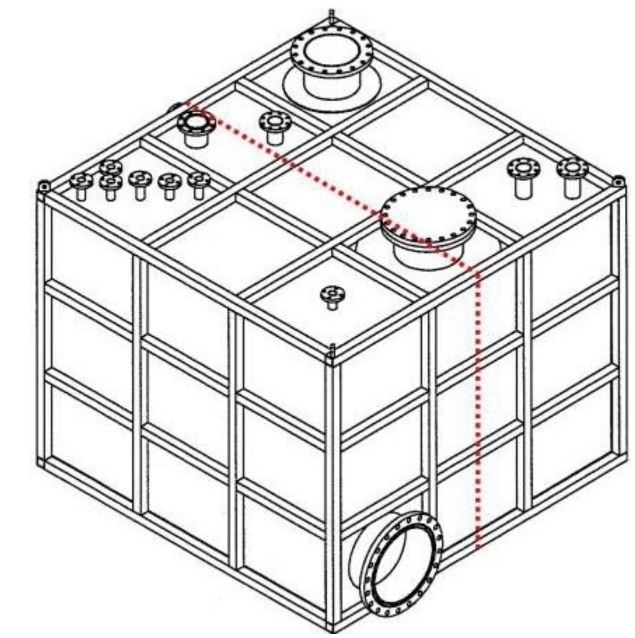
AN EXISTING TANK, THE 851 HI-RES TANK WILL BE A MIRRORED COPY.

IkC:SS OIHCRVISC SPCCIFIED DIMINSIONS MIE IM INCHES
TOLOW.CCSIJM
r AAC:IIONS C! S ANGLCS
± I/ 8 ' ±.! ± 2•
MATERIAL
S EE MATERIAL UST

B51 HI- RES TANK DETAILS
MICRON, BOISE IDAHO
MATERIALS LIST

SCALE• N ONE
SHEET: I OF 5

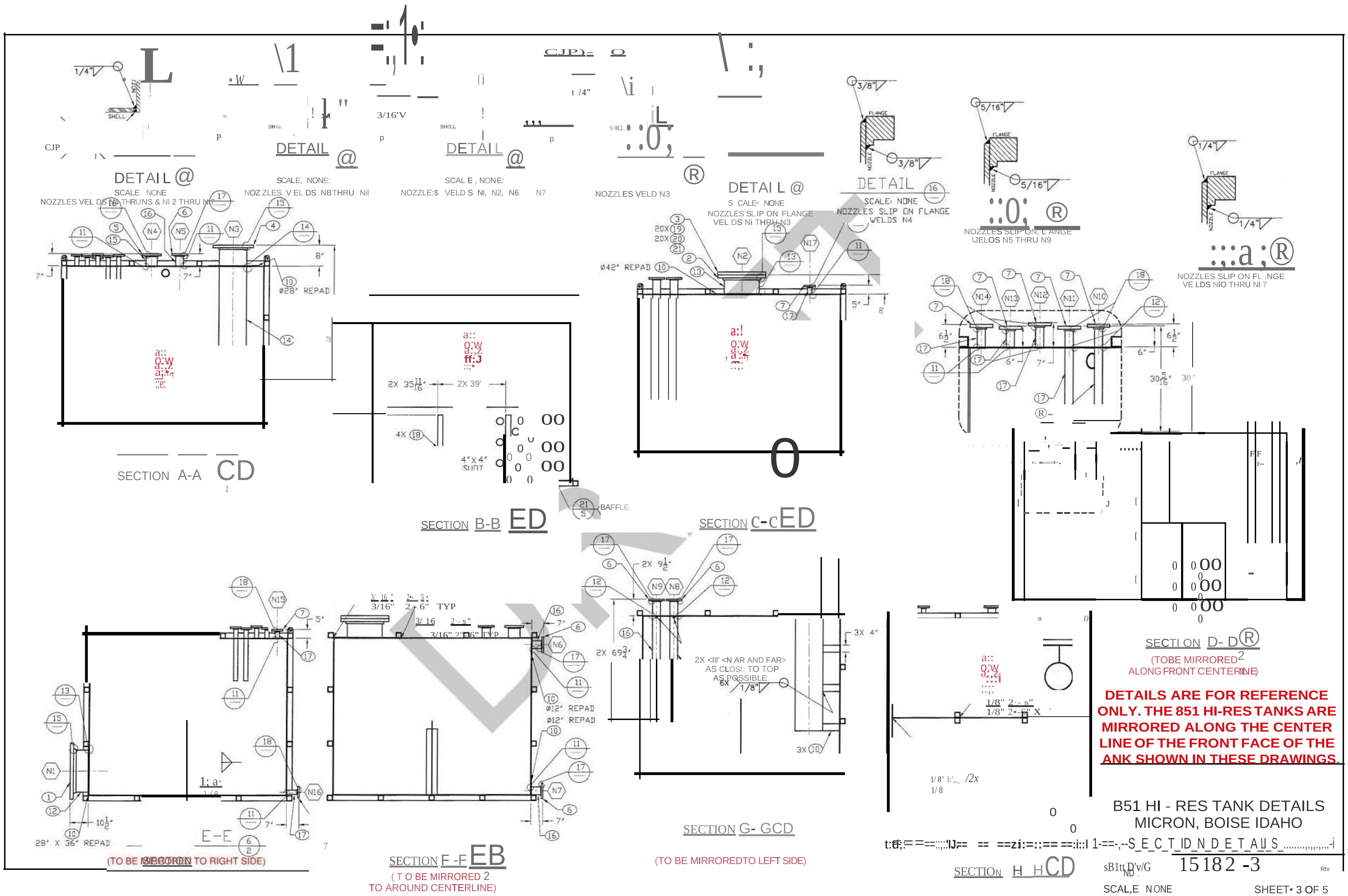




ISOMETRIC VIEW
(TO BE MIRRORED ALONG CENTER LINE OF FRONT FACE)

DETAILS ARE FOR REFERENCE ONLY. THE 851 HI-RES TANKS ARE MIRRORED ALONG THE CENTER LINE OF THE FRONT FACE OF THE TANK SHOWN IN THESE DRAWINGS.

B51 HI-RES TANK DETAILS MICRON, BOISE IDAHO OVERALL TANK PLANS	
SIZE DWG NO. 15182-2	REV
SCALE: NONE	SHEET: 2 OF 5



DETAILS ARE FOR REFERENCE ONLY. THE 851 HI-RES TANKS ARE MIRRORED ALONG THE CENTER LINE OF THE FRONT FACE OF THE TANK SHOWN IN THESE DRAWINGS.

B51 HI - RES TANK DETAILS

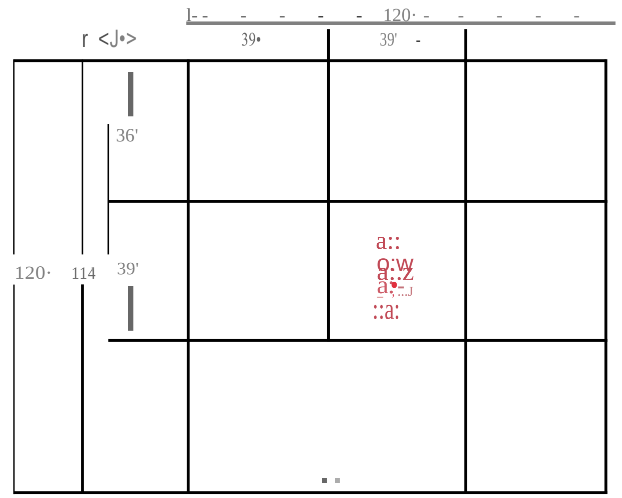
MICRON, BOISE IDAHO

SECTION DETAILS

15182 -3

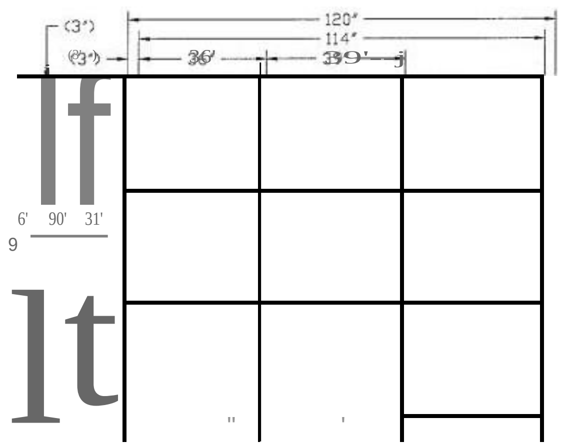
SCALE NONE

SHEET 3 OF 5



TOP VIE\J

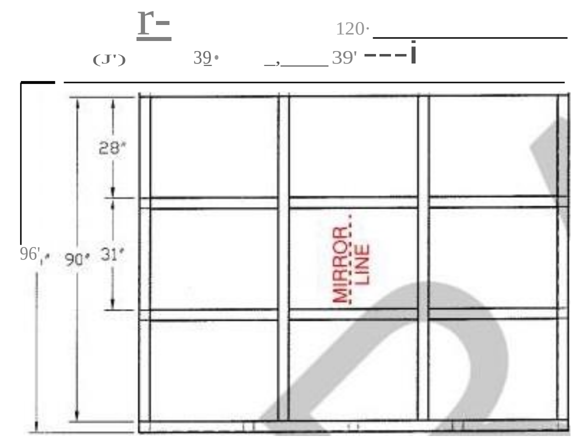
BOTTOM TUBE NOT SHO\JN FOR CLARIT Y



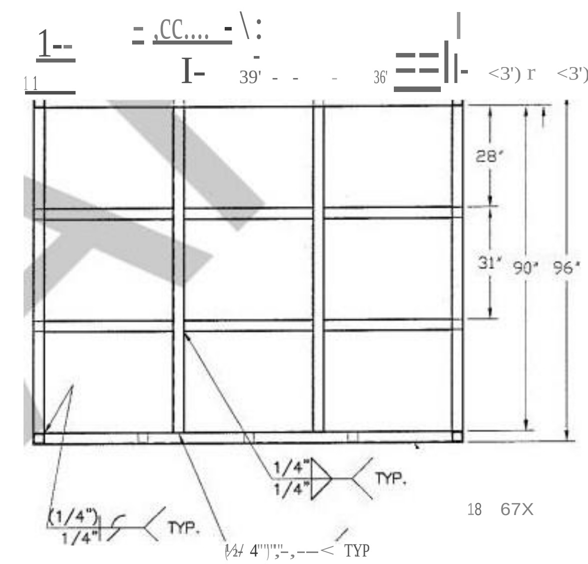
LEFT SIDE VIE\J

(TO BE MIRRORED TO RIGHT SIDE)

END CAP

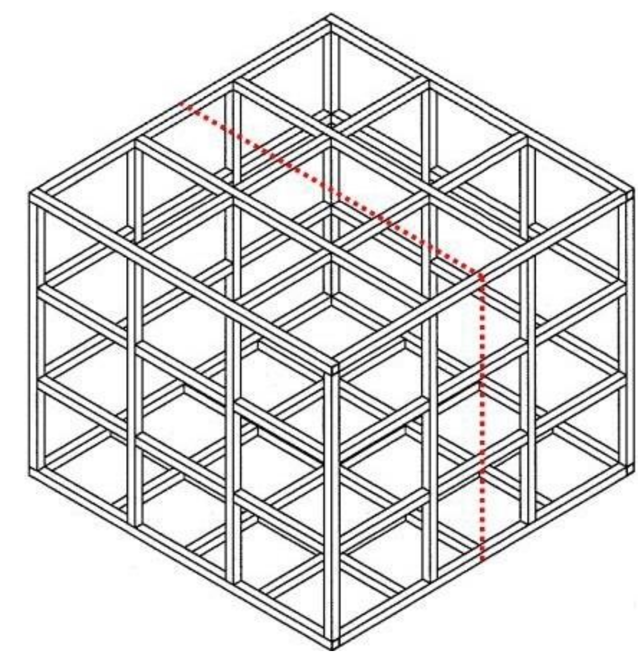


FRONT VIE\J



RIG HT S I DE VIE\J

(TO BE MIRRORED TO LEFTSIDE)



ISOMETRIC VIE\J

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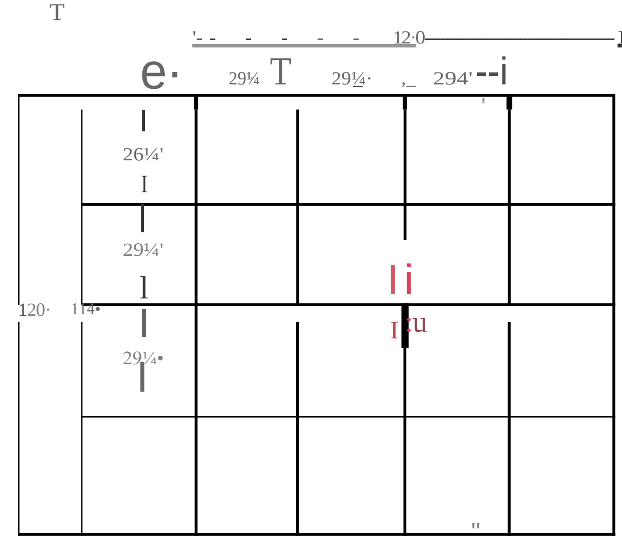


BOTTOM VIE\J

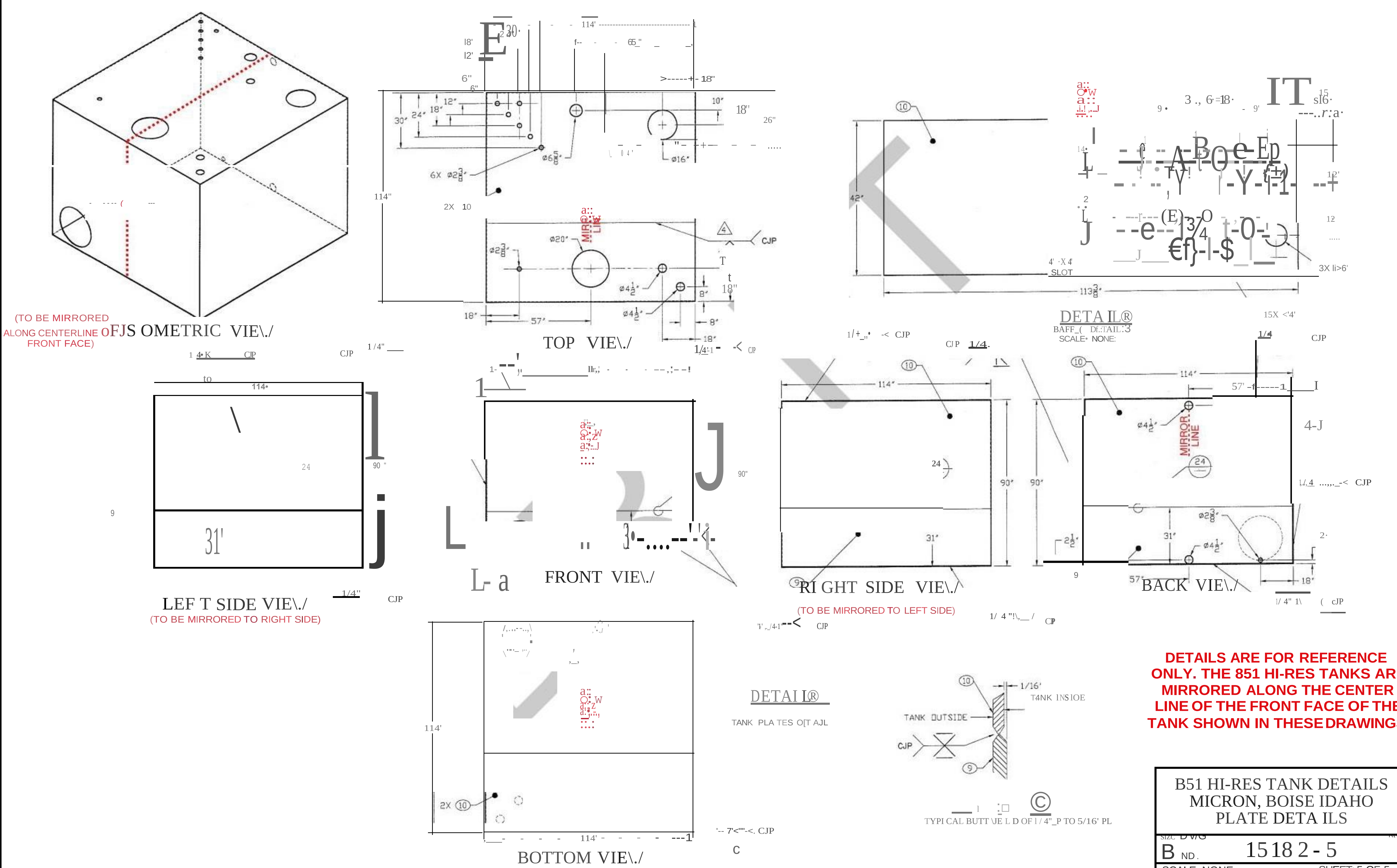
TOP TUBE NOT SHO\JN FOR CLARI TY

DETAIL(V-

TuBE SUPPORT DETA[l

DETAILS ARE FOR REFERENCE ONLY. THE 851 HI-RES TANKS ARE MIRRORED ALONG THE CENTER LINE OF THE FRONT FACE OF THE TANK SHOWN IN THESE DRAWINGS.

B51 HI-RES TANK DETAILS MICRON, BOIS E IDAHO SUPPORT DETA ILS	
SIZE B	DWG ND. 1518 2 - 4
SCALE• NONE	SHE E T: 4 OF 5



DETAILS ARE FOR REFERENCE ONLY. THE 851 HI-RES TANKS ARE MIRRORED ALONG THE CENTER LINE OF THE FRONT FACE OF THE TANK SHOWN IN THESE DRAWINGS.

B51 HI-RES TANK DETAILS
MICRON, BOISE IDAHO
PLATE DETAILS

SIZE B | DWG NO. 15182-5
SCALE: NONE | SHEET: 5 OF 5